Yelp nrtSearch: Disaggregated Search Architecture

Primary/Replica Ingestion Decoupling, gRPC Segment Streaming & S3 Archival

1. Real-Time Attribute Freshness at Yelp Scale

At Yelp, millions of consumers execute geo-spatial search queries for local businesses, restaurants, and professional services. However, critical metadata changes by the second:

The Operational Challenge

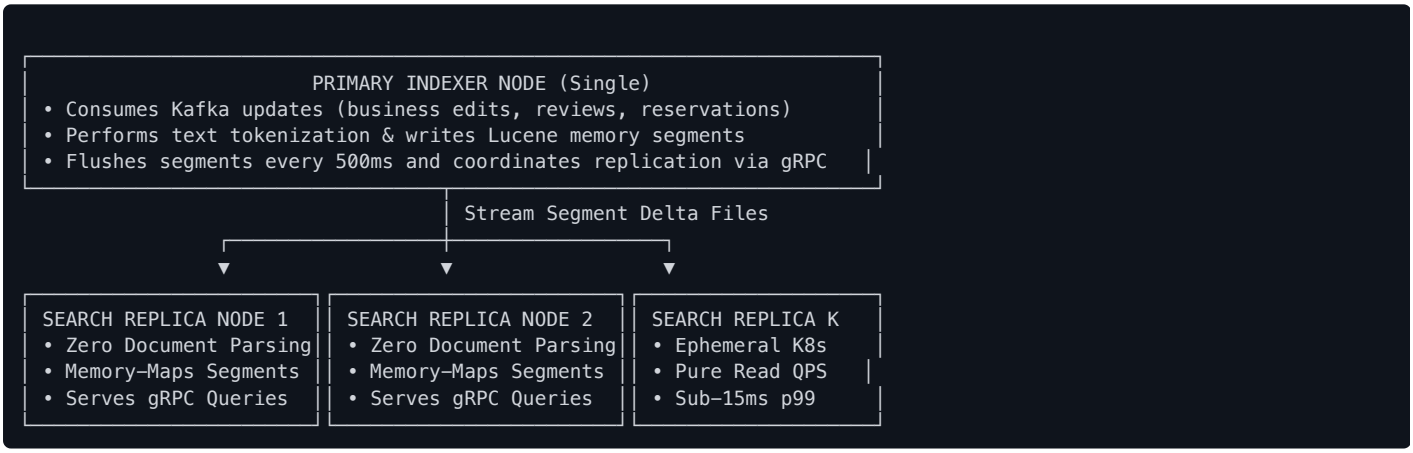
- **Instant Operational Status:** A business marking itself "Closed" or modifying emergency hours must reflect in search results within seconds to avoid misguiding users.
- **Real-Time Seat & Waitlist Capacity:** Live dining availability changes dynamically as reservations confirm or cancel.
- **High Query Concurrency:** Sustaining tens of thousands of complex geo-spatial, multi-facet queries per second with strict sub-30ms p99 latency SLAs.

2. Elasticsearch's Critical Bottlenecks

Yelp historically operated Elasticsearch. At production scale, its monolithic model failed:

- **Duplicate Document Parsing ($N \times$ CPU Waste):** In Elasticsearch, every replica independently parses JSON, runs text tokenizers, and builds Lucene segments. For 5 replicas, CPU work is multiplied by 5x.
- **Cluster State Churn:** Frequent mapping updates and node heartbeats saturated Elasticsearch master nodes, causing cluster-wide state lock freezes.
- **I/O Contention:** Background Lucene segment merging on read replicas saturated SSD bandwidth, causing p99 search latency to spike over 800ms.

3. The nrtSearch Architectural Topology



4. Physical Segment-Level Replication Mechanics

nrtSearch completely redefines search replication:

- **Index Once, Distribute Everywhere:** The Primary node alone performs compute-heavy tokenization, analysis, and indexing.
- **gRPC Segment Delta Transfer:** When a 500ms NRT commit occurs, the Primary notifies replicas of new segment filenames. Replicas pull only the compressed binary files over high-speed internal networks.
- **Zero-Parse Instant Visibility:** Replicas execute `IndexReader.openIfChanged()` over memory-mapped files, making newly indexed records searchable in **under 1 second** with negligible replica CPU overhead.

5. S3 Cloud Storage Disaggregation & Autoscaling

By persisting Lucene segments directly to **Amazon S3**, nrtSearch decouples storage durability from compute nodes:

Ephemeral Replica Autoscaling
During evening restaurant search peaks, Yelp spins up hundreds of stateless replica pods. Replicas hydrate local NVMe caches directly from S3 snapshots, joining active traffic in under 60 seconds.

Rapid Primary Failover
If a Primary crashes, a standby node mounts the S3 segment manifest, replays lagging Kafka offsets, and assumes Primary duties within seconds without data loss.

6. Architectural Comparison: Elasticsearch vs. nrtSearch

System Feature	Elasticsearch / OpenSearch	Yelp nrtSearch
Replication Granularity	Document-level (JSON re-indexing on all nodes)	**Segment-level (Pre-built Lucene binary files)**
Replica CPU Overhead	High (Duplicates 100% of indexing work)	**Zero indexing CPU (60% fleet reduction)**
Document Freshness SLA	10 – 60 seconds (Heavy merge pauses)	**Sub-second (< 500ms NRT commit)**
Network Interface	HTTP / JSON REST API	High-speed binary gRPC over HTTP/2

gRPC Scoring Plugins & Production Reliability Rules

Dynamic Ranking Models, Protocol Efficiency & Lessons

7. High-Performance gRPC Scoring & ML Rankers

Yelp replaces REST APIs with binary gRPC interfaces to maximize query throughput:

- **Protobuf Payloads:** Compact binary wire representation reduces network packet serialization time by 40% compared to bulky JSON.
- **Native Collector Plugins:** Search requests execute custom scoring algorithms (BM25 relevance, geo-distance decay, user review sentiment) directly within the Lucene low-level hit collector loop, bypassing memory allocations.

8. Key Architectural Takeaways for Search Platforms

Core Principles Distilled from Yelp nrtSearch

- **Stream Pre-Built Segments Instead of Raw Documents:** Shifting from document-level replication to physical segment streaming eliminates redundant CPU tokenization across the replica fleet, cutting compute costs by 60%.
- **Isolate Ingestion from Search Workloads:** Decoupling the write-primary from read replicas prevents heavy Lucene segment compaction from degrading read query latencies.
- **Leverage Object Storage for Stateless Autoscaling:** Persisting segment archives to S3 transforms replica nodes into ephemeral, easily scalable compute instances that bootstrap in seconds.
- **Binary gRPC Trumps JSON for Internal Microservices:** High-throughput search federations benefit dramatically from gRPC multiplexing and binary Protobuf schemas.